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Storage system including battery module and method for operating the same

Abstract

A storage system includes a host, a back plane including a switching block and connected to the host, a plurality of storage sets connected to the back plane to receive power from the host, and a battery module connected to the back plane to supply spare power to at least one storage set of the plurality of storage sets through the switching block, when an abnormal power drop is detected in the at least one storage set of the plurality of storage sets. The battery module controls the at least one storage set to perform a data flushing operation, when power less than or equal to a threshold value is detected in the at least one storage set after supplying the spare power.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application claims priority under 35 U.S.C. § 119 to Korean Patent Application Nos. 10-2022-0067031 filed on May 31, 2022 and 10-2022-0097524 filed on Aug. 4, 2022 in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.
BACKGROUND
(2) Embodiments of the present disclosure described herein relate to a storage system including a battery module and a method of operating the same.
(3) A storage device is a device that stores data under the control of a host, such as a computer, a smartphone, or a smart pad, and includes a device that stores data in a semiconductor memory, especially a non-volatile memory, such as a solid state drive (SSD) or a memory card.
(4) The storage device may perform a computing function, depending on implementing aspects, and may include an additional volatile memory and a core for the computing function such that the computing function may be performed. The storage device may operate by receiving power, and during the operation, a Sudden Power Off (SPO) situation may occur, where power is suddenly cut off. In this case, the storage device may perform a data flushing operation that dumps data, which is stored in the volatile memory, to the non-volatile memory. When the volatile memory has a larger size, an amount of data to be dumped may be increased. In this case, there is a limitation in supplying power for dumping by using only the capacity of the battery additionally provided in the storage device.
SUMMARY
(5) Embodiments of the present disclosure provide a storage system including a battery module, capable of preventing data loss, when a power drop occurs in a storage device performing a computing function and a method of operating the same.
(6) According to example embodiments, a storage system includes a host, a back plane including a switching block and connected to the host, a plurality of storage sets connected to the back plane to receive power from the host, and a battery module connected to the back plane to supply spare power to at least one storage set of the plurality of storage sets through the switching block, when an abnormal power drop is detected in the at least one storage set of the plurality of storage sets. The battery module controls the at least one storage set to perform a data flushing operation, when power less than or equal to a threshold value is detected in the at least one storage set after supplying the spare power.
(7) According to example embodiments, a storage system may include a host, a back plane including a switching block, and connected to the host, a plurality of storage sets connected to the back plane to receive power from the host, and a battery module connected to the back plane to be offloaded a power management operation for the plurality of storage sets from the host, and performs the power management operation through the switching block.

(8) According to example embodiments, an operating method of a storage system may include monitoring whether an abnormal power drop is detected in at least one storage set of a plurality of storage sets included in the storage system, supplying spare power to the at least one storage set when the power drop is detected, and controlling the at least one storage set to perform a data flushing operation, when power less than or equal to a threshold value is detected in the at least one storage set after supplying the spare power.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) The above and other objects and features of the present disclosure will become apparent by describing in detail embodiments thereof with reference to the accompanying drawings.
(2) FIG. **1** illustrates a storage system according to an embodiment of the present disclosure;
(3) FIG. **2** illustrates a host according to an embodiment of the present disclosure;
(4) FIGS. **3** and **4** are views illustrating an operation for supplying power of a battery module according to various embodiments of the present disclosure;
(5) FIGS. **5** and **6** are views illustrating a storage set according to various embodiments of the present disclosure;
(6) FIG. **7** is a view illustrating a battery module according to an embodiment of the present disclosure;
(7) FIG. **8** is a flowchart illustrating a method of operating a battery controller according to an embodiment of the present disclosure;
(8) FIG. **9** is a flowchart illustrating a method of operating a battery controller according to another embodiment of the present disclosure;
(9) FIGS. **10**A and **10**B are views illustrating an operating of managing power of a battery controller according to example embodiments;
(10) FIG. **11** illustrates a method of operating a storage system including a battery controller according to an embodiment of the present disclosure;
(11) FIG. **12** illustrates a battery module according to another embodiment of the present disclosure; and
(12) FIG. **13** is a flowchart illustrating a method of operating a storage system including a battery module of FIG. **12** according to example embodiments.
DETAILED DESCRIPTION
(13) Hereinafter, embodiments of the present disclosure will be described clearly and in detail such that those skilled in the art may easily reproduce the present disclosure.
(14) FIG. **1** illustrates a storage system according to an embodiment of the present disclosure.
(15) Referring to FIG. **1**, a storage system **100** according to an embodiment of the present disclosure includes a host **120**, a back plane **140**, a storage set **160**, and a battery module **180**.
(16) The storage system **100** may be implemented, for example, in the form of a server, a data center, a personal computer (PC), a network-coupled storage, an Internet of Things (IoT) device, or an electronic device. The electronic device may be one of a laptop computer, a mobile phone, a smartphone, a tablet PC, a personal digital assistant (PDA), an enterprise digital assistant (EDA), a digital still camera, a digital video camera, an audio device, a portable multimedia player (PMP), a personal navigation device (PND), an MP3 player, a handheld game console, an e-book, and a wearable device.
(17) The host **120** may communicate with the storage set **160** through various interfaces. For example, the host **120** may be implemented in the form of an application processor (AP) or a system-on-a-chip (SoC). In addition, for example, the host **120** may be implemented in the form of an integrated circuit or a main board, but the present disclosure is not limited thereto.

(18) The host **120** may transmit a write request and a read request to the storage set **160**. The storage set **160** may store data received from a host, in response to the write request, or may read the stored data, in response to the read request and transmit the read data to the host **120**.
(19) In addition, the host **120** may control the use of the power of the storage set **160**. For example, the host **120** may perform an operation of balancing the load of power or control a power management operation, such as an operation of supplying power to the storage set **160** through the back plane **140**, monitoring power, or supplying, by the battery module **180**, a spare power to the storage set **160** under a situation (for example, the SPO or power glitch) of abnormally supplying power. Alternatively, for example, the host **120** may offload the power management operation to the battery module **180**, such that the battery module **180** performs the power management operation.
(20) The back plane **140** may be provided between the host **120** and the storage set **160** and may be connected to the host **120** and the storage set **160**. The back plane **140** may be configured to exchange data through various communication protocols.
(21) The back plane **140** includes a switching block **141** configured to selectively supply power to at least one storage set of the storage set **160**. The switching block **141** may be controlled by the host **120** or the battery module **180** and may provide power to the storage set **160** which needs to be supplied with power depending on a switching operation.
(22) The storage set **160** and the battery module **180** may be received in a form factor. The form factor may include various form factors complying with a standard specification, and for example, may, but is not limited to, include an Enterprise and Data Center Standard Form Factor, such as E3.S, E3.S2T, E3.L, and E3.L2T.
(23) The storage set **160** may include one storage set or a plurality of storage sets **160***a* to **160***c*, and may be connected to the host **120** and the battery module **180** through the back plane **140**. Herein, for convenience of description, the terms of the storage set **160** and the plurality of storage sets **160** may be used interchangeably. Each of the plurality of storage sets **160***a* to **160***c* is a device having a computing function and a data storage function, and may also be referred to as a smart storage device (e.g., smartSSD). The storage set **160** may receive power from the host **120** and the battery module **180** through the back plane **140**.
(24) The battery module **180** is connected to the host **120** and the plurality of storage sets **160** through the back plane **140**.
(25) According to an embodiment, the battery module **180** may be configured to be offloaded a power management operation for the plurality of storage sets **160** from the host **120** and to perform the power management operation through the switching block **141**. For example, at least a portion of the power management operation of the host **120** may be offloaded to the battery module **180**. In this case, the battery module **180** may perform at least a portion of the power management operation, in place of the host **120**.
(26) According to an embodiment, the battery module **180** may be configured to detect an abnormal power drop in at least one storage set of the plurality of storage sets **160** and to supply the spare power through the switching block **141**, when the power drop is detected. The abnormal power drop may refer to a situation, such as the SPO or the power glitch described above, in which power of the storage set **160** is rapidly lowered.
(27) According to an embodiment, the battery module **180** may control at least one storage set to perform a data flushing operation, when power less than or equal to a threshold value is detected from the at least one storage set after supplying the spare power. According to the present disclosure, the data flushing operation may be defined as an operation of dumping data stored in the volatile memory into the non-volatile memory. For example, the battery module **180** may control the at least one storage set to dump data, which is stored in the volatile memory, to the non-volatile memory. For example, at least one storage set may include the volatile memory and the non-volatile memory.

(28) According to embodiments as described above, the storage system **100** may further include the battery module **180** capable of offloading the power management operation of the host **120** to prevent the loss of data stored in the storage set **160** even in the situation, such as SPO or power glitch, in which the power is abnormally detected. According to the present disclosure, the increased power amount of the storage set **160** may be covered by adding a computing function to the storage set **160**, instead of additionally including a power loss protection (PLP) block or a PLP battery to cope with the abnormal power detection.
(29) Hereinafter, components included in the storage system **100** will be described.
(30) FIG. **2** illustrates a host, according to an embodiment of the present disclosure.
(31) Referring to FIG. **2**, the host **120** according to an embodiment of the present disclosure includes a power supply **121** and a main board **122**. The power supply **121** generates power PWR from a power source and supplies the generated power PWR to the main board **122**. For example, the main board **122** may supply power PWR to the back plane **140**. Alternatively, the power supply **121** may directly supply power PWR to the back plane **140**.
(32) The main board **122** may be referred to as a mother board or a base board, and includes a first processor **123**, a plurality of first memories **124***a* and **124***b* connected to the first processor **123**, a second processor **125**, a plurality of second memories **126***a* and **126***b* connected to the second processor **125**, and a baseboard management controller (BMC) **127**.
(33) The first processor **123** may use the plurality of first memories **124***a* and **124***b* as operating memories, and the second processor **125** may use the plurality of second memories **126***a* and **126***b* as operating memories. The first processor **123** and the second processor **125** may be configured to execute an operating system and various applications.
(34) The first processor **123** and the second processor **125** may access the back plane **140** to control the use of the power for the plurality of storage sets **160**. For example, the first processor **123** and the second processor **125** may perform an operation of balancing the load of power, or may control a power management operation, such as an operation of supplying power to the storage set **160** through the back plane **140**, monitoring power, or supplying, by the battery module **180**, the spare power to the storage set **160** under a situation (for example, the SPO or power glitch) of abnormally supplying power. Alternatively, for example, the first processor **123** and the second processor **125** may offload the power management operation to the battery module **180** such that the battery module **180** performs the power management operation.
(35) For example, the first processor **123** and the second processor **125** may be a central processing unit CPU, and the plurality of first memories **124***a* and **124***b* and the plurality of second memories **126***a* and **126***b* may be volatile memories, such as dynamic random access memory (DRAM) or static RAM (SRAM).
(36) The BMC **127** may be a system separate from the first processor **123** and the second processor **125**, and may monitor the physical state, such as a temperature, humidity, a voltage of the power supply **121**, a fan speed, a communication parameter, or an operating system function, of components of the storage system **100**. Alternatively, for example, the BMC **127** may offload the power management operation to the battery module **180**.
(37) FIGS. **3** and **4** are views illustrating an operation for supplying power of a battery module according to various embodiments of the present disclosure.
(38) Referring to FIG. **3**, the back plane **140** may receive power directly or indirectly from the power supply **121** included in the host **120** and may transmit the power, which is supplied, to the plurality of storage sets **160** through the switching block **141** of the back plane **140**.
(39) The back plane **140** is connected to the host **120** (e.g., BMC **127**, the first processor **123**, and the second processor **125**), each storage set of the plurality of storage sets **160**, and the battery module **180** through connection interfaces CI_CB, CI_PB, CI_BS, and CI_BB, respectively. The connection interface may be an interface protocol, for example, a Peripheral Component Interconnect-Express (PCI-E), Advanced Technology Attachment (ATA), Serial ATA (SATA),

Parallel ATA (PATA), serial attached SCSI (SAS) or Computer eXpress Link (CXL), or another interface protocol such as a System Management (SM) bus, a Universal Serial Bus (USB), a Multi-Media Card (MMC), an Enhanced Small Disk Interface (ESDI), or an Integrated Drive Electronics (IDE).

(40) The switching block **141** may be connected to each of the plurality of storage sets **160** through the above-described various connection interfaces CI_BS, and the battery module **180** may control the switching block **141** through the connection interface CI_BB to supply power to at least one storage set.

(41) Referring to FIG. **4**, the connection interface according to an embodiment may include a CXL interface (CXL_E), a PCIe interface (PCIe_E), and an SM bus (SM_BUS). In this case, the BMC **127** of the host **120** is connected to the back plane **140** through the SM bus (SM_BUS), and at least one of the first processor **123** and the second processor **125** is connected to the back plane **140** through the CXL interface (CXL_E) and the PCIe interface (PCIe_E). The plurality of storage sets **160** are connected to the back plane **140** through the CXL interface (CXL_E), the PCIe interface (PCIe_E), and the SM bus (SM_BUS). Alternatively, according to an embodiment, the CXL interface (CXL_E) may not be provided. In this case, the host **120** may also be able to access data using CXL protocols (e.g., CXL.io, CXL, cache, and CXL memory) through the PCIe interface (PCIe_E).

(42) The switching block **141** is connected to each of the plurality of storage sets **160** through the PCIe interface (PCIe_E), and is switched under the control of the battery module **180** to supply power to at least one storage set.

(43) FIGS. **5** and **6** illustrate a storage set according to various embodiments of the present disclosure.

(44) Referring to FIG. **5**, a storage set **160_1** according to an embodiment is connected to the back plane **140** through a connection interface CI_BS and includes an accelerator **161**, a buffer memory **162**, a storage device **170**, a power loss protection (PLP) block **165**, and a PLP battery **166**. Herein, the storage set **160_1** may be one of the plurality of storage sets **160***a* to **160***c*.

(45) The accelerator **161** may perform an acceleration function to assist the computation of the host **120** by performing some of the operations performed by the host **120**. For example, the accelerator **161** may be connected to the storage device **170** to receive input data from the storage device **170**, to perform a computation for the input data such that computation data is generated, and to store the generated computation data in a buffer memory or transmit the computation data to the storage device **170**. The accelerator **161** may perform the above-described computing operation, in response to a command of the host **120**.

(46) The buffer memory **162** may store the computation data based on the computation from the accelerator **161**. Alternatively, according to an embodiment, the buffer memory **162** may store system data offloaded from the host **120**. The buffer memory **162** may be a volatile memory such as DRAM or SRAM.

(47) The storage device **170** may be an internal memory embedded in the electronic device. For example, the storage device **170** may be a solid state drive (SSD), an embedded universal flash storage (UFS) memory device, or an embedded multi-media card (eMMC). According to some embodiments, the storage device **170** may be an external memory detachable to or from the electronic device. For example, the storage device **170** may be a UFS memory card, Compact Flash (CF), Secure Digital (SD), Micro-SD (MicroSecure Digital), Mini Secure Digital (Mini-SD), extreme digital (xD) or a Memory Stick.

(48) The storage device **170** may communicate with the host **120** through the connection interface CI_BS described above. The story device **170** may receive a command (host command) from the host **120** and analyze the command to generate a command for controlling the accelerator **161**.

(49) The storage device **170** includes a storage controller **171**, a buffer memory **172**, and a non-volatile memory (NVM) **173**. For example, the storage controller **171**, the buffer memory **172**, and

the NVM **173** may be implemented in the form of individual chips, respectively. The storage controller **171** may generate input data necessary for the host **120** to perform the requested operation, based on a command, or may read the input data out of the buffer memory **172** or the NVM **173**.
(50) The PLP block **165** is included in the storage device **170** to prevent power loss in an abnormal power drop situation. The PLP block **165** is connected to the PLP battery **166** and may perform an operation of supplying power to the storage device **170** to prevent data loss, when a power drop is detected while monitoring the power of the storage device **170**. However, since the PLP battery **166** uses a capacitor, it may be difficult to supply a sufficient amount of power to the storage device **170**.
(51) Referring to FIG. **6**, although a storage set **160_2** according to another embodiment is connected to the back plane **140** through the connection interface CI_BS, and includes the accelerator **161**, the buffer memory **162**, and the storage device **170**, the PLP block **165** and the PLP battery **166** may be omitted. Herein, the storage set **160_2** may be one of the plurality of storage sets **160***a* to **160***c*. When the PLP block **165** and the PLP battery **166** are omitted, the battery module **180** performs a function of the PLP block **165**. For example, when the abnormal power drop is detected in the storage device **170**, the storage device **170** may receive power from the battery module **180** and may perform the data flushing operation that dumps data stored in the buffer memory **162** or **172** to the NVM **173** under the control of the battery module **180**.
(52) FIG. **7** illustrates a battery module according to one embodiment of the present disclosure.
(53) Referring to FIG. **7**, a battery module **180_1** includes a battery controller **181** and at least one battery **182**. Herein, the battery module **180_1** of FIG. **7** may be replaced with the battery module **180** of FIGS. **1**, **3**, and **4**.
(54) The battery controller **181** is connected to the back plane **140** through the connection interface CI_BB, offloaded the power management operation for the plurality of storage sets **160** from the host **120**, and performs the offloaded power management operation through the switching block **141**. The battery controller **181** may be a logic chip which may be reconfigured and reprogrammed for offloading. For example, the battery controller **181** may be a field programmable gate array (FPGA), a microcontroller unit (MCU), a programmable logic device (PLD), or a complex PLD (CPLD).
(55) At least one battery **182** is electrically connected to the battery controller **181**, to supply power stored therein to the storage set **160** under the control of the battery controller **181**. The at least one battery **182** may be configured to store energy greater than that of the PLP battery **166** in the storage device **170**.
(56) Hereinafter, various embodiments related to the battery controller **181** of FIG. **7** will be described.
(57) Power Management Operation
(58) According to an embodiment, the battery controller **181** may be offloaded the power management operation of at least one of the BMC **127**, the first processor **123** and the second processor **125** included in the host **120** through the connection interface CI_BB. Alternatively, the battery controller **181** may be offloaded not only the power management operation but also information, such as a power policy or a priority set for each of the plurality of storage sets **160**, for each of the plurality of storage sets **160** through the connection interface (CI_BB). The priority may be set be higher for the storage set **160** which needs to be supplied with power first, when an abnormal situation (e.g., SPO or power glitch) occurs. In this case, the storage set **160** having a high need for receiving power may be, for example, the storage set **160** which performs a computation having a higher need for protection or stores computation data resulting from the computation.
(59) For example, the battery controller **181** may be offloaded power management operations based on the TP 4091 standard implemented such that NVM express (NVMe) supports offloading the

function of the host **120** to the storage device **170**.
(60) FIG. **8** is a flowchart illustrating an operating method of a battery controller, according to an embodiment of the present disclosure.
(61) Referring to FIG. **8**, in step S**110**, the battery controller **181** may be offloaded at least one of the power management operation for the plurality of storage sets **160**, the power policy set for each of the plurality of storage sets **160**, and the priority set for each of the plurality of storage sets **160**. For example, by programming, the battery controller **181** may be offloaded at least one of the power management operation for the plurality of storage sets **160**, the power policy set and the priority set for each of the plurality of storage sets **160**. For example, the host **120** may offload at least one of the power management operation for the plurality of storage sets **160**, the power policy and the priority for each of the plurality of storage sets **160**, to the battery controller **181**.
(62) In step S**120**, the battery controller **181** monitors the power of the storage set **160** through the connection interface. The monitoring of the power is an operation for sensing the abnormal power drop described above. For example, the monitoring of the power for the storage set **160** may be performed through an SM bus (SM_BUS).
(63) In step S**130**, the battery controller **181** supplies spare power to the storage set **160**, based on at least one of the power management operation, the power policy, or the priority, which is offloaded, when the abnormal power drop occurs. For example, the battery controller **181** may first provide spare power to the storage set **160** having the highest priority or at least one storage set having the higher priority, based on the priorities set for the plurality of storage sets **160**.
(64) According to an embodiment, the battery controller **181** may perform an operation of balancing the load of power. The battery controller **181** monitors power of the storage set **160** through the connection interface CI_BB. The battery controller **181** detects a work load for each of the plurality of storage sets **160** through monitoring. For example, the work load may refer to a load for each storage set **160** depending on an input/output operation or a computation operation between the host **120** and the storage set **160**.
(65) The battery controller **181** sets a threshold value for each of the plurality of storage sets **160** based on the work load. In this case, the threshold value refers to a power threshold for performing the above-described data flushing. For example, the battery controller **181** may set a higher threshold for one storage set **160**, when it is determined that an amount of computation performed or to be performed in the one storage set **160** is greater than an amount of computation performed or to be performed in another storage set **160**. For example, the battery controller **181** may set a higher threshold for any one storage set **160** of the plurality of storage sets **160**, when it is determined that an amount of data to be written to or read out of the one storage set **160** of the plurality of storage sets **160** is greater than an amount of data written to or read out of another storage set **160** of the plurality of storage sets **160**.
(66) In addition, the battery controller **181** sets an amount of spare power to be supplied for each of the plurality of storage sets **160**, based on the work load, when the power drop occurs. For example, the storage set **160** having a larger amount of work load may be set to have a larger amount of spare power.
(67) FIG. **9** is a flowchart illustrating an operation of a battery controller, according to another embodiment of the present disclosure.
(68) Referring to FIG. **9**, in step S**210**, the battery controller **181** monitors and detects a work load for the plurality of storage sets **160**.
(69) In step S**220**, the battery controller **181** determines at least one of a threshold value or a spare power for each of the plurality of storage sets **160**, based on the detected work load. For example, the battery controller **181** may set the threshold value or the spare power to be higher, as the work load is increased as described above.
(70) In step S**230**, the battery controller **181** determines whether the abnormal power drop is detected in at least one storage set of the plurality of storage sets **160** through the monitoring. When

the abnormal power drop is not detected in S**230** (e.g., No), the battery controller **181** may perform step S**210**.
(71) When it is determined that the power drop is detected in step S**230**, the battery controller **181** starts supplying power based on the spare power determined by using at least one battery **182** in step S**240**.
(72) When the power drop consecutively occurs even if the spare power is supplied, the battery controller **181** may perform the data flushing operation based on the threshold value in step S**250**.
(73) According to embodiments of the present disclosure as described above, the battery controller **181** may be offloaded the power management operation from the host **120** and perform at least portion of the power management operation or a power supplying operation of the plurality of storage sets **160**, in place of the host **120**, thereby improving the efficiency of the storage system **100**. In addition, the battery controller **181** may set the threshold value or the spare power necessary for the power management operation by individually considering the work load for each of the plurality of storage sets **160**.
(74) Power Drop Protection
(75) According to an embodiment, the battery controller **181** monitors the plurality of storage sets **160** and may detect abnormal power drops from the plurality of storage sets **160** through monitoring. For example, the battery controller **181** may monitor the storage set **160** through the SM bus SM_BUS included in the connection interface.
(76) When a power drop is detected, the battery controller **181** controls the switching block to supply the spare power, which is received from at least one battery **182**, to at least one storage set where the power drop is detected. The battery controller **181** controls at least one storage set to perform a data flushing operation, when power less than or equal to the threshold value is detected in at least one storage set after supplying the spare power.
(77) FIGS. **10**A and **10**B are graphs illustrating a power management operation of a battery controller according to example embodiments.
(78) Referring to FIG. **10**A, before a power drop from the plurality of storage sets **160** is detected, that is, when the power drop is not detected, the plurality of storage sets **160** performs an operation of supplying power from the host **120**. When an abnormal power drop is detected from the storage set **160** at a time point of t**1**, the spare power is started to be supplied. For example, the spare power may be supplied immediately at the time point of t**1** when the power drop is detected, or may be supplied at a time point of t**2** after a preset time from the time point of t**1** when the power drop is detected. When the power less than or equal to the threshold value is not detected in at least one storage set after supplying the spare power, the battery controller **181** may continue to monitor power drop again.
(79) Referring to FIG. **10**B, when the power drop is detected at the time point of t**1**, the battery controller **181** similarly starts supplying the spare power at the time point of t**2**. When power less than or equal to the threshold value is detected due to the consecutive power drop even though the spare power is supplied at a time point of t**3**, the battery controller **181** controls the storage set **160** to perform the data flushing operation. For example, the battery controller **181** controls the storage set **160** to perform the data flushing operation at the time point of t**3** when the power (e.g., Pth) less than or equal to the threshold value is detected, and the storage set **160** dumps data, which is stored in the buffer memory, to the NVM.
(80) FIG. **11** illustrates a method of operating a storage system including a battery controller, according to an embodiment of the present disclosure.
(81) Referring to FIG. **11**, in step S**310**, the host **120** included in the storage system **100** first offloads at least one of a power management operation for the plurality of storage sets **160** or a power policy or a priority set for each of the plurality of storage sets **160**. The offloading operation may be performed through the connection interface CI_BB. To this end, the battery controller **181** included in the battery module **180_1** may be reconfigured and reprogrammed.

(82) In step S**320**, the storage set **160** performs normal write/read operations R/W, based on the power supplied from the supplied general power.
(83) In step S**330**, the battery module **180_1** monitors power of the storage set **160**. The operation of monitoring the power is an operation for detecting the above-described abnormal power drop. For example, the operation of monitoring power for the storage set **160** may be performed through the SM bus (SM_BUS).
(84) In step S**340**, the battery module **180_1** determines whether abnormal power, i.e., power drop, is detected during the operation of monitoring the power. When the power drop is not detected (e.g., No), the battery module **180_1** may consecutively perform step S**330**.
(85) When a power drop is detected in step S**340**, the battery module **180_1** supplies the spare power to the storage set **160** in step S**350**. The battery module **180_1** may supply the spare power to at least one storage set, in which a power drop is detected, of the plurality of storage sets **160** through the switching block **141** included in the back plane **140**.
(86) In step S**360**, the battery module **180_1** monitors whether the power of at least one storage set is less than or equal to the threshold value (Pth) while supplying the spare power. When the power is not less than or equal to the threshold value (Pth), it may be determined that the power is recovered by supplying the spare power, and thus step S**320** and step S**330** may be performed again.
(87) When the power of at least one storage set is detected as being less than or equal to the threshold (Pth) in step S**360**, the battery module **180_1** controls at least one storage set to perform a data flushing operation in step S**370**. The at least one storage set dumps data, which is stored in the volatile memory, to the NVM to prevent data from being lost due to the power drop.
(88) According to the above-described embodiments of the present disclosure, especially when a power glitch, which is a temporary power drop phenomenon, is detected, the battery module **180_1** immediately supplies the spare power. When the power is lowered to be less than or equal to the threshold value, the battery module **180_1** may control the storage set **160** to perform the data flushing, thereby preventing data from being lost even under a power glitch situation.
(89) Data Copy
(90) According to an embodiment, the battery controller **181** may copy data of the storage set **160** to cope with when an abnormal power drop is detected. To this end, the battery controller **181** may be programmed to copy data. Hereinafter, an embodiment related to a data copy function will be described with reference to FIGS. **11** and **12**.
(91) FIG. **12** illustrates a battery module according to another embodiment of the present disclosure.
(92) Referring to FIG. **12**, a battery module **180_2** according to another embodiment may include a buffer memory **183** and an NVM **184** in addition to the above-described battery controller **181** and at least one battery **182**. Herein, the battery module **180_2** of FIG. **12** may be replaced with the battery module **180** of FIGS. **1**, **3**, and **4**. The buffer memory **183** and the NVM **184** may store the copied data, when the data of the storage set **160** is copied.
(93) The battery controller **181** monitors the plurality of storage sets **160** and may detect abnormal power drops from the plurality of storage sets **160** through monitoring. For example, the battery controller **181** may monitor the storage set **160** through the SM bus (SM_BUS) included in the connection interface CI_BB. Alternatively, the battery controller **181** may determine power consumption (that is, usual power consumption) of the plurality of storage sets **160** when the power drop is not detected through the monitoring operation.
(94) When a power drop is detected through monitoring, the battery controller **181** determines whether to select the storage set **160**, in which the power drop is detected, as an alternative storage set, based on at least one of the usual power consumption or the set priority of the storage set **160** in which the power drop is detected. For example, when the storage set **160** shows lower usual power consumption (e.g., when the usual power consumption is less than or equal to the preset

power consumption), the battery controller **181** may determine the risk of the failure of data copy as being lowered in the storage set **160** and may select the storage set **160** as an alternative storage set.

(95) When selecting the storage set **160** in which the power drop is detected as the alternative storage set, the battery controller **181** copies data stored in the non-volatile memory included in the storage set **160** in which the power drop occurs, through the connection interface CI_BB. When data copy is completed, the battery controller **181** notifies the host **120** of information on the data copy completion. Thereafter, when receiving the information on the data copy completion, the host **120** reads the copied data stored in the battery module **180**.

(96) FIG. **13** is a flowchart illustrating a method of operating a storage system including the battery module of FIG. **12** according to example embodiments.

(97) Referring to FIG. **13**, in step S**410**, the host **120** included in the storage system **100** first offloads at least one of a power management operation for the plurality of storage sets **160** or a power policy or a priority set for each of the plurality of storage sets **160**. The offloading operation may be performed through the connection interface. To this end, the battery controller **181** included in the battery module **180_2** may be reconfigured and reprogrammed. In particular, information on the priority required for selecting an alternative storage set may be offloaded.

(98) In step S**420**, the battery module **180_2** monitors power of the storage set **160**. The operation of monitoring the power is an operation for detecting the above-described abnormal power drop or determining the usual power consumption. For example, the operation of monitoring the power for the storage set **160** may be performed through the SM bus (SM_BUS).

(99) In step S**430**, the battery module **180_2** determines whether abnormal power, i.e., power drop, is detected while monitoring the power. When the power drop is not detected, the battery module **180_2** may continuously perform step S**420**.

(100) When a power drop is detected in step S**430**, the battery module **180_2** determines, in step S**440**, whether to select the storage set **160**, in which the power drop is detected, as an alternative storage set, based on at least one of the usual power consumption or the set priority of the storage set **160** in which the power drop is detected.

(101) When the storage set **160** in which the power drop is detected is selected as the alternative storage set, the battery module **180_2** copies data stored in the alternative storage set in step S**450**. The copied data may be stored in the memory (the buffer memory **183** and/or the NVM **184**) included in the battery module **180_2**.

(102) In step S**460**, when data copy is completed, the battery module **180_2** notifies the host **120** of information on the data copy completion.

(103) In step S**470**, the host **120** reads the coped data from the battery module **180_2**.

(104) According to the embodiments of the present disclosure, when a power drop is detected, an alternative storage set may be selected based on the priority and the usual power consumption, and data from the selected alternative storage set is copied to prevent the data loss.

(105) As described above, the present disclosure may provide a storage system including a battery module, capable of preventing data loss when a power drop occurs in a storage device performing a computing function and a method for operating the same.

(106) The above description refers to detailed embodiments for carrying out the present disclosure. Embodiments in which a design is changed simply or which are easily changed may be included in the present disclosure as well as an embodiment described above. In addition, technologies that are easily changed and implemented by using the above embodiments may be included in the present disclosure. While the present disclosure has been described with reference to embodiments thereof, it will be apparent to those of ordinary skill in the art that various changes and modifications may be made thereto without departing from the spirit and scope of the present disclosure as set forth in the following claims.

(107) While the present disclosure has been described with reference to embodiments thereof, it

will be apparent to those of ordinary skill in the art that various changes and modifications may be made thereto without departing from the spirit and scope of the present disclosure as set forth in the following claims.

Claims

1. A storage system comprising: a host; a back plane including a switching block and connected to the host; a plurality of storage sets connected to the back plane and configured to receive power from the host; and a battery module connected to the back plane and configured to: supply spare power to at least one storage set of the plurality of storage sets through the switching block, in response to an abnormal power drop being detected in the at least one storage set of the plurality of storage sets, and control the at least one storage set to perform a data flushing operation, in response to the power from the host less than or equal to a threshold value being detected in the at least one storage set after supplying the spare power, wherein the abnormal power drop is detected when the power from the host drops to a first value greater than the threshold value, wherein the battery module is configured to: detect the first value of the power greater than the threshold value at a first time point and a second value of the power less than or equal to the threshold value at a second time point after the first time point, supply the spare power to the at least one storage set in response to the power being detected the first value at the first time point, and control the at least one storage set to perform the data flushing operation in response to the power being detected the second value at the second time point, and wherein the second value of the power is detected at the second time point when the power drops from the first value at the first time point to the second value at the second time point.
2. The storage system of claim 1, wherein each of the plurality of storage sets includes: a storage device including a non-volatile memory; an accelerator connected to the storage device and configured to perform a computation in response to a command from the host; and a volatile memory configured to store at least one of computation data, which results from the computation and system data offloaded from the host.
3. The storage system of claim 2, wherein the data flushing operation is defined as an operation of dumping the computation data, which is stored in the volatile memory, into the non-volatile memory.
4. The storage system of claim 1, wherein the battery module includes: at least one battery; and a battery controller configured to detect the abnormal power drop and control the switching block such that the spare power is supplied to the at least one storage set from the at least one battery.
5. The storage system of claim 4, wherein the battery controller is further configured to: detect a work load for each of the plurality of storage sets, and set the threshold value for each of the plurality of storage sets, based on the work load.
6. The storage system of claim 4, wherein the battery controller is further configured to: detect a work load for each of the plurality of storage sets, and set the spare power for each of the plurality of storage sets, based on the work load.
7. The storage system of claim 1, further comprising: a PCIe interface and a system management bus configured to connect the back plane to the plurality of storage sets.
8. The storage system of claim 7, wherein the battery module is configured to supply the spare power to the at least one storage set through the PCIe interface.
9. The storage system of claim 7, wherein the battery module is configured to detect the abnormal power drop from the plurality of storage sets through the system management bus.
10. The storage system of claim 1, wherein the battery module is further configured to continue monitoring the abnormal power drop in response to the power less than or equal to the threshold value not being detected in the at least one storage set after supplying the spare power.
11. A storage system comprising: a host; a back plane connected to the host and including a

switching block; a plurality of storage sets connected to the back plane and configured to receive power from the host; and a battery module connected to the back plane, including at least one battery, and configured to: be offloaded a power management operation for the plurality of storage sets from the host, and to perform the power management operation through the switching block, detect whether an abnormal power drop occurs in at least one storage set of the plurality of storage sets, supply spare power to at least one storage set of the plurality of storage sets from the at least one battery in response to the abnormal power drop being detected, and control the at least one storage set to perform data flushing in response to the power from the host less than or equal to a threshold value being detected in the at least one storage set after supplying the spare power, wherein the abnormal power drop is detected when the power from the host drops to a first value greater than the threshold value, wherein the battery module is configured to: detect the first value of the power greater than the threshold value at a first time point and a second value of the power less than or equal to the threshold value at a second time point after the first time point, supply the spare power to the at least one storage set in response to the power being detected the first value at the first time point, and control the at least one storage set to perform the data flushing in response to the power being detected the second value at the second time point, and wherein the second value of the power is detected at the second time point when the power drops from the first value at the first time point to the second value at the second time point.
12. The storage system of claim 11, wherein the battery module includes: a battery controller configured to perform the power management operation, based on at least one of a power policy or a priority set for each of the plurality of storage sets.
13. The storage system of claim 11, wherein the battery module includes a battery controller configured to: detect a work load for each of the plurality of storage sets, and set the threshold value for each of the plurality of storage sets, based on the work load.
14. The storage system of claim 11, wherein the battery module includes a battery controller configured to: detect a work load for each of the plurality of storage sets, and set the spare power for each of the plurality of storage sets, based on the work load.
15. The storage system of claim 11, further comprising: a PCIe interface and a system management bus configured to connect the back plane to the plurality of storage sets, wherein the battery module is configured to detect the abnormal power drop from the plurality of storage sets through the system management bus, and supply power through the PCIe interface.
16. An operating method of a storage system, the operating method comprising: supplying power from a host to a plurality of storage sets included in the storage system; monitoring whether an abnormal power drop is detected in at least one storage set of the plurality of storage sets; supplying spare power to the at least one storage set when the abnormal power drop is detected; and controlling the at least one storage set to perform data flushing, when the power from the host less than or equal to a threshold value is detected in the at least one storage set after supplying the spare power, wherein the monitoring of whether the abnormal power drop is detected includes detecting the abnormal power drop when the power from the host drops to a first value greater than the threshold value, wherein the detecting of the abnormal power drop includes detecting the first value of the power greater than the threshold value at a first time point, wherein the controlling of the at least one storage set to perform the data flushing includes detecting a second value of the power less than or equal to the threshold value at a second time point after the first time point, and wherein the second value of the power is detected at the second time point when the power drops from the first value at the first time point to the second value at the second time point.
17. The operating method of claim 16, further comprising: detecting a work load for each of the plurality of storage sets; and setting the threshold value for each of the plurality of storage sets, based on the work load.
18. The operating method of claim 17, further comprising: setting the spare power for each of the plurality of storage sets, based on the work load.